Why Quantum Computing?

A short history, the course structure and logistics

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History at a glance: four eras, three threads

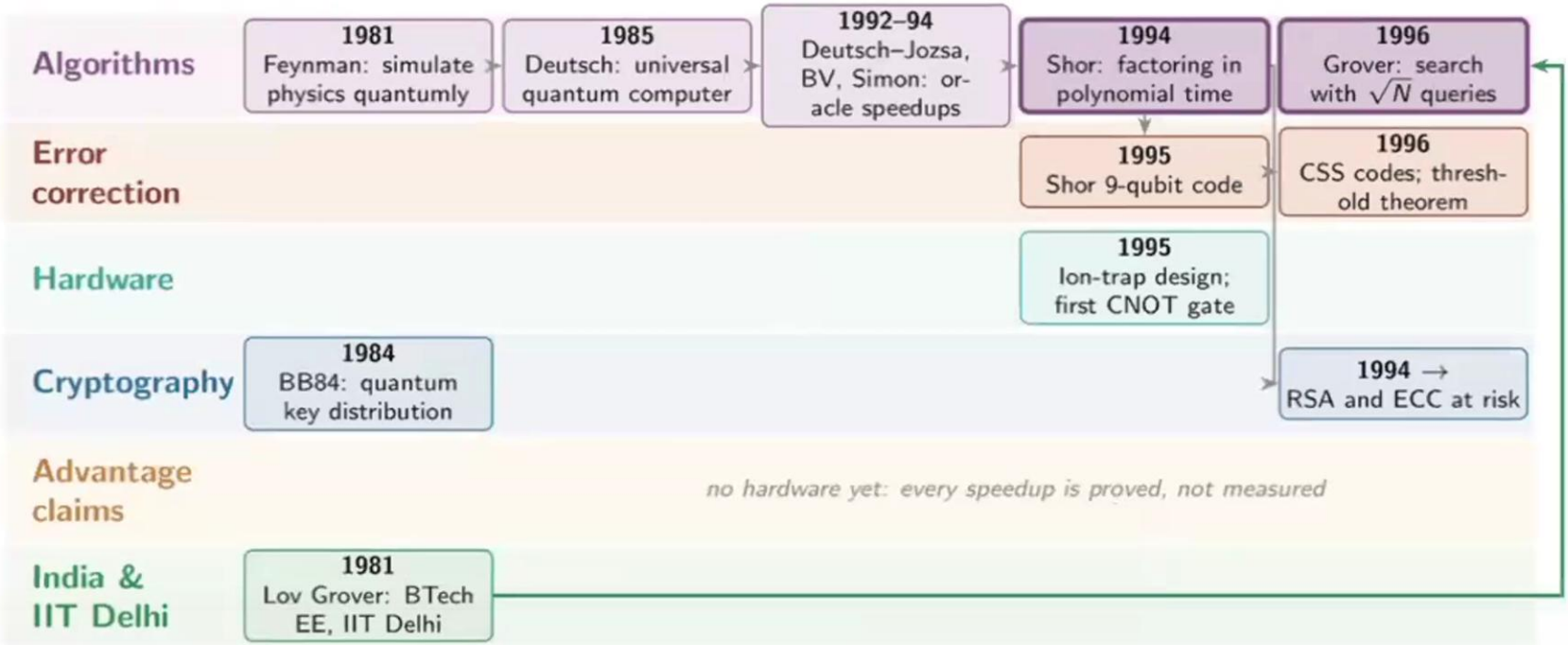


Algorithms: what a quantum computer *could* do (Shor 1994 → resource estimates today)

Error correction: whether it can be built at all (Shor code 1995 → Willow 2024)

Advantage claims: whether it already beats classical (D-Wave 2011 → Quantum Echoes 2025)

Era 1 (1980–1996): the ideas



Three Key Moments

Feynman (1981): "Nature is not classical, dammit" — proposed using quantum systems to simulate quantum physics, planting the seed for everything below.

Deutsch (1985)

First proof that a quantum computer can solve a problem *in one query* that requires two classically.

Establishes the **quantum circuit model** and the concept of quantum parallelism.

"Can a quantum computer do something a classical one cannot?" — answered: **yes**.

Shor (1994)

Factors an n-bit integer in $O(n^3)$ quantum gates — **polynomial time**, vs. subexponential time for the best known classical algorithms.

Threatens RSA and ECC on a sufficiently large fault-tolerant quantum computer; directly motivates post-quantum cryptography (NIST 2024 standards).

The moment the field became impossible to ignore.

Grover (1996) ★ IITD alumnus

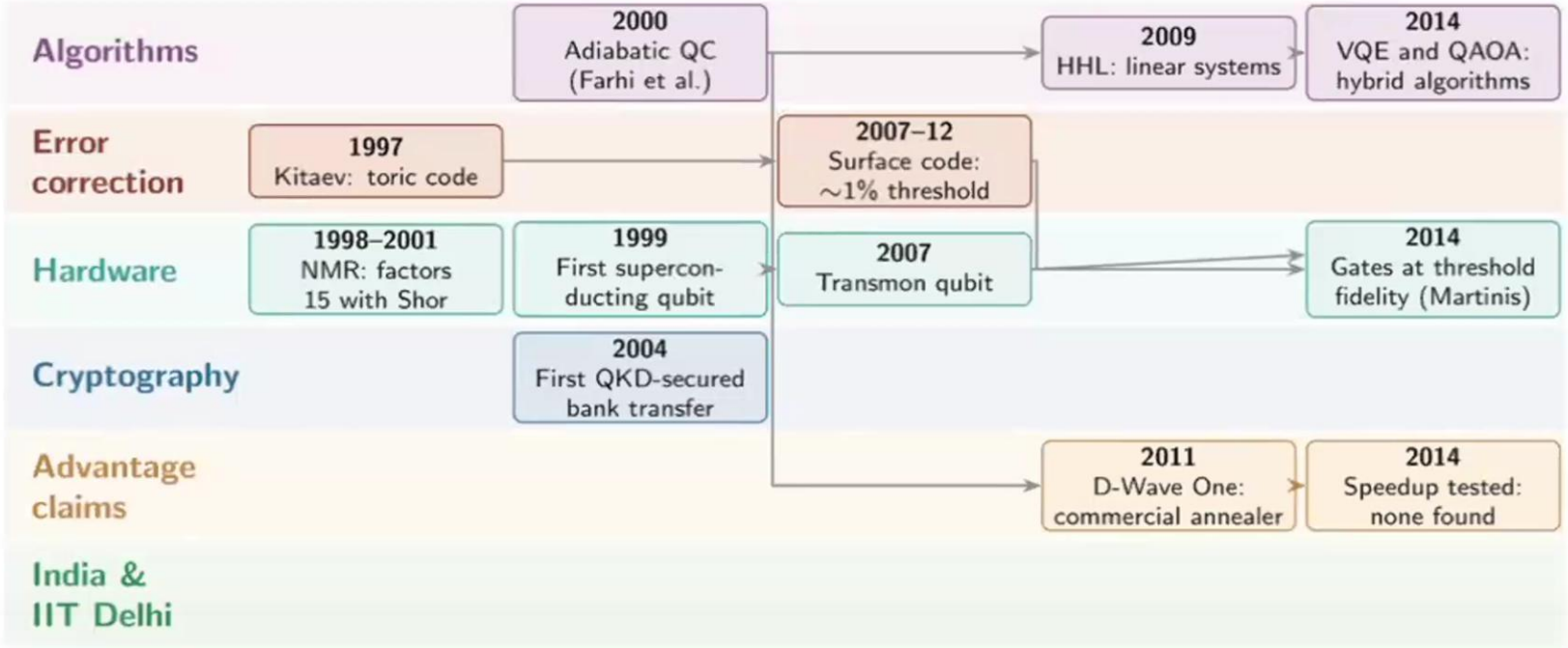
Searches N items with $\approx \frac{\pi}{4}\sqrt{N}$ queries; provably **optimal**.

Lov Grover — BTech EE, **IIT Delhi (1981)**; Bell Labs. IITD Distinguished Alumni Award, 2021.

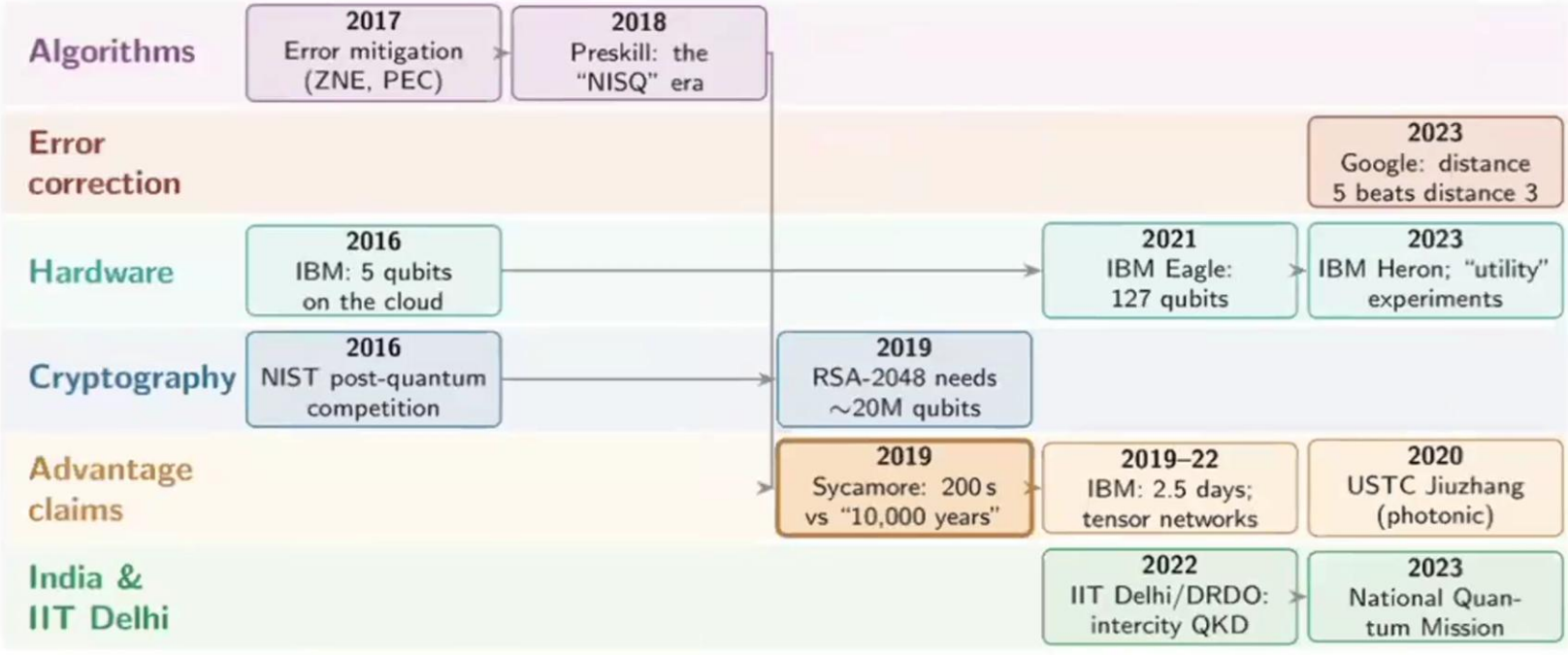
N	Grover vs. $N/2$
10^6	785 vs. 500k
10^9	24,800 vs. 500M

All three results are proved, not measured — quantum advantage as a mathematical theorem.

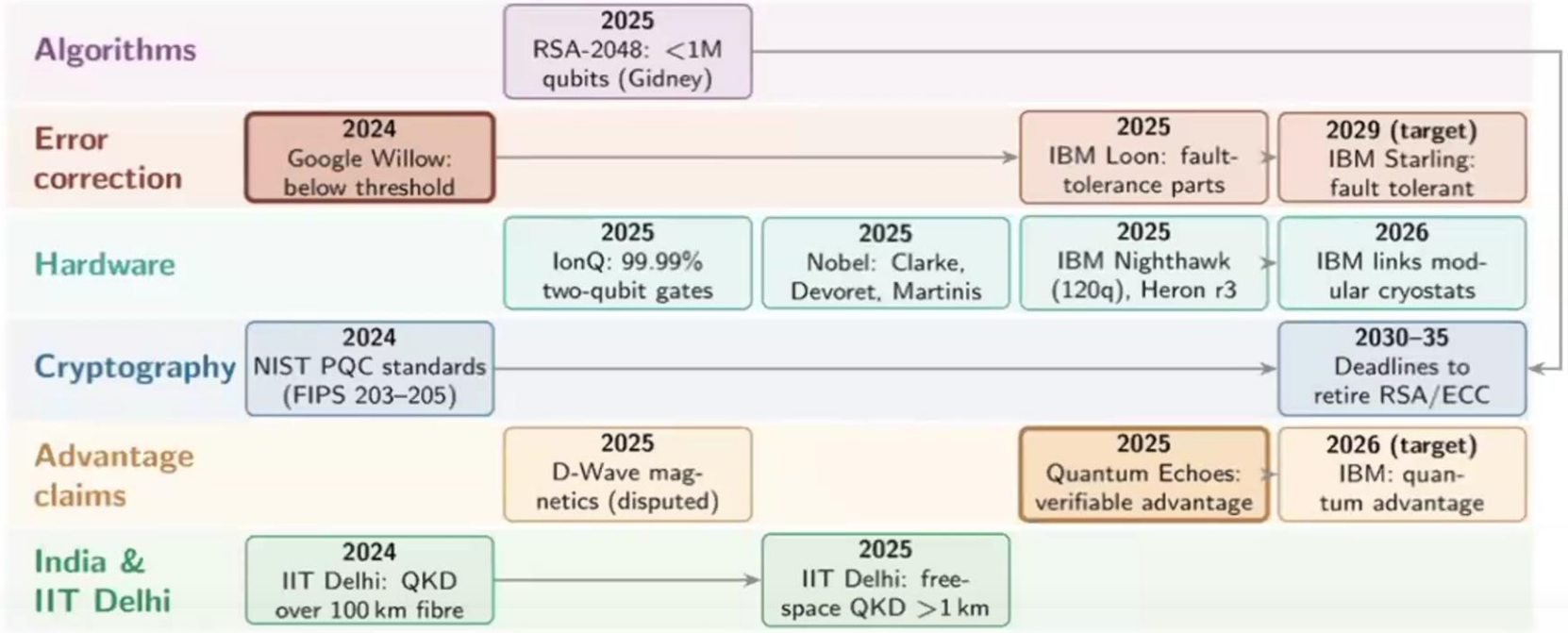
Era 2 (1997–2015): the first qubits



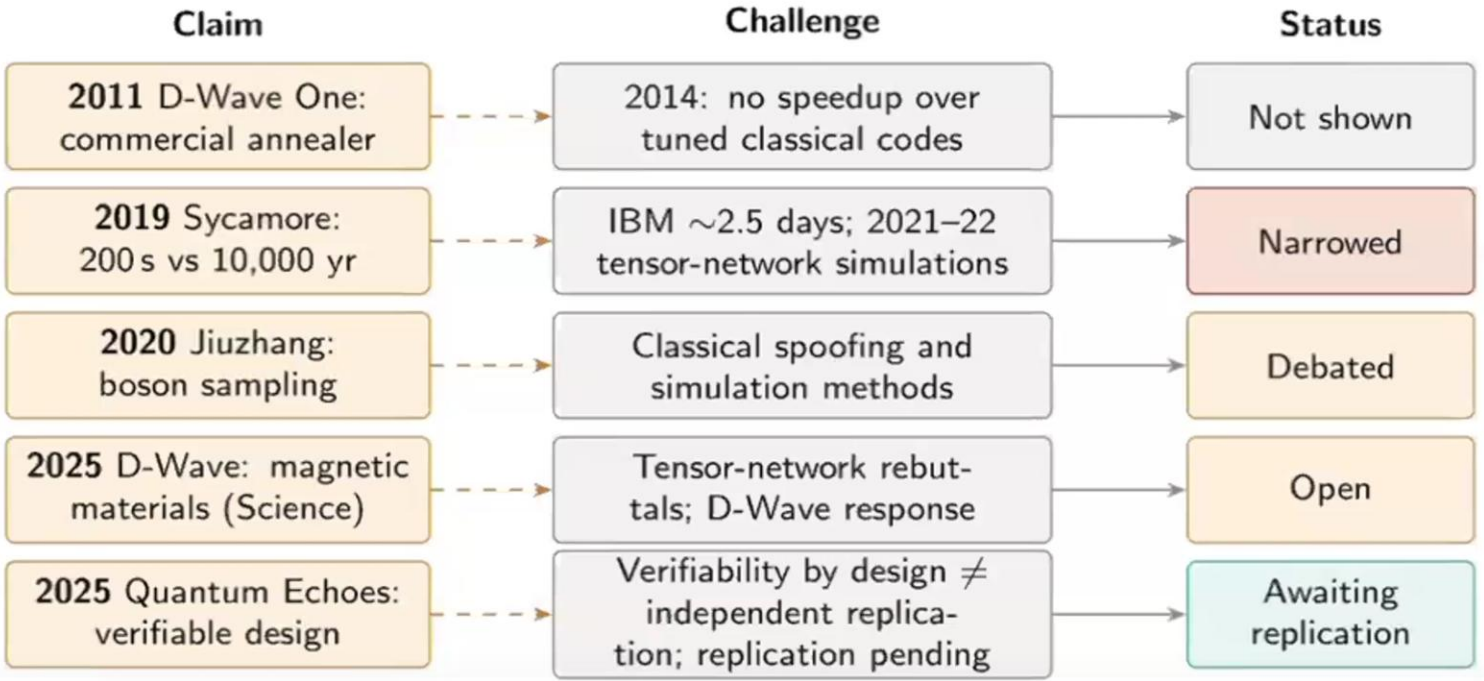
Era 3 (2016–2023): NISQ and "supremacy"



Era 4 (2024–2026): error correction starts to work



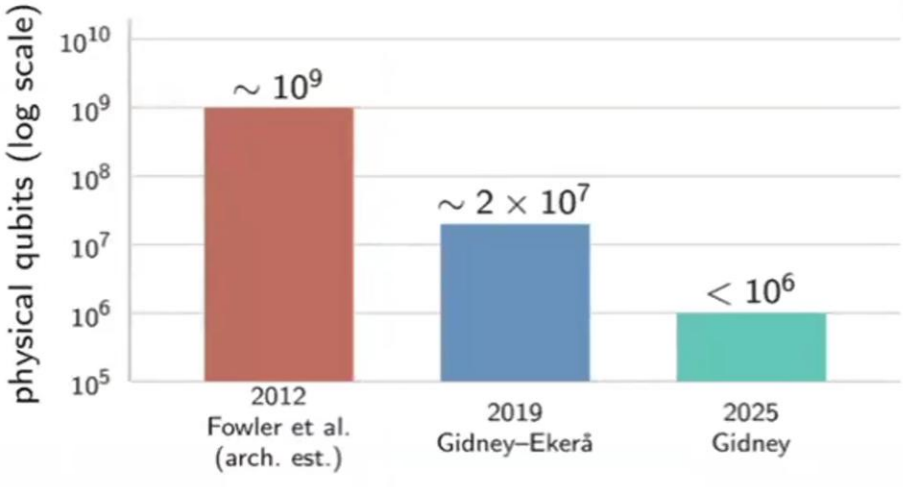
Advantage claims and their pushback



Lesson: an advantage claim is a moving target — classical algorithms improve too. In this course you will learn to judge one yourself

The cost of breaking RSA-2048 keeps falling

Each bar is a **theoretical resource estimate** *(not a demonstrated capability) for factoring a 2048-bit RSA key on a fault-tolerant quantum computer; bars reflect different algorithmic approaches, each superseding the last. Hardware roadmap targets are aspirational.*



- Same target, over **1000× fewer qubits** in 13 years
- Gains came from **algorithms, codes and compilation**, not only hardware
- Hardware roadmaps aim at 10^4–10^6 qubits by 2028–2030

[1] Fowler et al., "Surface codes: towards practical large-scale quantum computation," *Phys. Rev. A* 86, 032324 (2012)

[2] Gidney et al., "How to factor 2048 bit RSA integers in 8 hours using 20 million noisy qubits," *Quantum* 5, 433 (2021)

[3] Gidney, "How to factor 2048 bit RSA integers with less than a million noisy qubits," arXiv:2505.15917 (2025)

Why the future looks promising

A key QEC milestone demonstrated

Logical error rates now *fall* as codes grow (Willow, 2024) — a necessary step toward fault tolerance, not the finish line.

Costs keep falling

Better algorithms and codes cut resource estimates by orders of magnitude.

Many platforms are racing

Superconducting, trapped ions, neutral atoms, photonics, spin qubits.

Roadmaps to fault tolerance

IBM Starling (2029), IonQ, Google, Quantinuum: logical qubits this decade.

Investment and policy

National missions, including India's NQM; PQC deadlines create demand now.

The opportunity

Quantum software, control systems, cryptography migration, research.

Quantum research at IIT Delhi: three active fronts

Semiconducting Qubit Hardware

Abhishek Dixit (EE) led group building a semiconducting qubit based Quantum Computer

Kolin Paul (CSE) building compiler stack and RFSoC-based control & readout electronics.

Some of the concepts for compilation and pulse-control topics are part of this Course.

Quantum Communication & QKD

Bhaskar Kanseri (Physics) — entangled photon sources, BB84/E91 implementations, free-space and fibre links.

Milestones:

- 2022 — intercity QKD (DRDO)
- 2024 — 100 km fibre QKD
- 2025 — free-space QKD >1 km

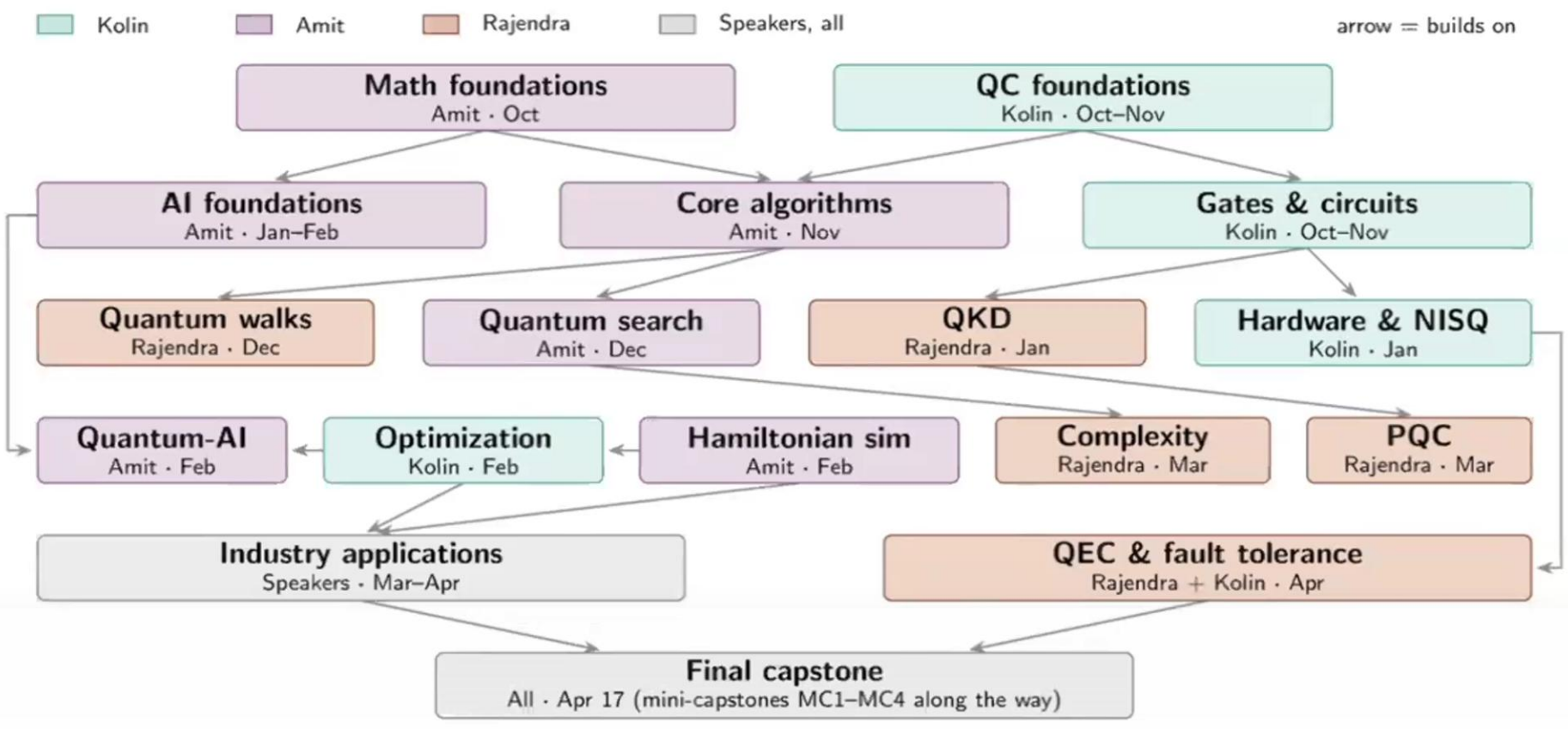
National Quantum Mission

India's NQM (Apr 2023):
INR 6003 Cr · 8 years · 4 T-Hubs

IITD is active across the hubs for quantum computing, communication, and sensing & metrology (atomic clocks, magnetometers, gravimeters).

Three NQM pillars — computing, communication, sensing — on one campus.

Course map: modules, dependencies, instructors



Five modules at a glance

M1 · Foundations	M2 · Core Algorithms	M3 · Circuits & Hardware	M4 · Complexity & QML	M5 · Crypto & QEC
Math + QC basics (Oct–Nov)	QFT, Shor, Grover, search (Nov–Dec)	Gates, NISQ, devices (Oct–Jan)	BQP/QMA, walks, AI (Feb–Mar)	QKD, PQC, fault tolerance (Jan–Apr)

Module	Topics	Instructor
M1 Foundations	Linear algebra, qubits, entanglement, gates	Amit + Kolin
M2 Core Algorithms	Deutsch–Jozsa, QFT, phase est., Shor, Grover	Amit
M3 Circuits & HW	Circuit model, universality, NISQ, devices	Kolin
M4 Complexity & AI	BQP/QMA, walks, Hamiltonian sim, quantum ML	Rajendra + Amit
M5 Crypto & QEC	QKD, post-quantum crypto, error correction, FT	Rajendra + Kolin

Logistics

Class Schedule

Saturday — Lectures (4h)

10:00 – 12:00	Block 1: lecture
12:00 – 12:30	Break
12:30 – 14:30	Block 2: lecture

Sunday — Problem Solving (2h)

11:30 – 13:30	Problem solving

Problems drawn from Saturday's content.
Attendance strongly recommended.

Self Study

Additional problems posted each week.
Not discussed in class – solutions posted 2 weeks later.

Mini-Capstones & Final Capstone

MC1	Circuit harness & noise characterisation
MC2	Hardware-aware compilation
MC3	QAOA starter + quantum kernel starter
MC4	Post-quantum cryptography (standalone)
Final	End-to-end hybrid application

Textbooks

Start here	N. D. Mermin, *Quantum Computer Science*, Cambridge 2007. Short, written for CS, no physics assumed.
Main reference	Nielsen & Chuang, *Quantum Computation and Quantum Information*, Cambridge 2010. The standard text; chapters map to modules.